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Annexe 2
Annex 2

Valeurs recommandées des fréquences étalons destinées à la mise en pratique de la définition du mètre et aux représentations secondaires de la seconde

Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second

Partie 2
Part 2

Calcium 40 Ion ()

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CCL-CCTF Frequency Standards Working Group

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Recommended values of standard frequencies for applications including the practical realization of the metre and secondary representations of the second Part 2: Calcium 40 Ion ()

CCL-CCTF Frequency Standards Working Group

9^{th} **edition** **2019**

1. Recommended value [1] of the frequency in the CIPM List of Frequencies [2]

$f\left(^{40}\mathrm{Ca}^{+}\right) = 411\ 042\ 129\ 776\ 399.8\ \mathrm{Hz}$

equivalent to

$\lambda\left(^{40}\mathrm{Ca}^{+}\right) = 729\ 347\ 276.793\ 944\ \mathrm{fm}$,

with a relative standard uncertainty of 2.4×10^{-15}.

2. Source data

The adopted value of $f(^{40}\mathrm{Ca}^{+})$ = 411 042 129 776 399.8 Hz with a relative standard uncertainty of 2.4×10^{-15}

has been calculated by the methods presented in [3], [4], [5], [6], with the data from the table of frequencies (2017) attached below. Since the frequency and the associated uncertainty of ref. [26] is not compatible with the other data, the uncertainty given in the original publication was increased to 3 Hz to make it statistically more consistent. This value has very little effect, and so the recommended frequency value comes essentially only from measurements made by two labs. Thus, the final uncertainty has been increased by a factor of two following the procedures outlined in ref. [6].

Table 1. Table of evaluated frequencies (2017)

Frequency / Hz	Uncertainty / Hz	Reference
^{199}Hg		
1 128 575 290 808 155.1	6.4	[7]
1 128 575 290 808 154.62	0.41	[8]
^{171}Yb		
518 295 836 590 864.0	28.0	[38]
518 295 836 590 863.1	2.0	[39]
518 295 836 590 865.2	0.7	[40]
518 295 836 590 863.5	8.1	[41]
518 295 836 590 863.59	0.31	[42]
518 295 836 590 863.38	0.57	[43]
^{88}Sr		
429 228 066 418 009.0	32.0	[9]
429 228 066 418 008.3	2.1	[10], Sr1
429 228 066 418 007.3	2.9	[10], Sr2
^{87}Sr		
429 228 004 229 874.0	1.1	[11]
429 228 004 229 873.65	0.37	[12]
429 228 004 229 873.6	1.1	[13]
429 228 004 229 874.1	2.4	[14]
429 228 004 229 872.9	0.5	[15]
429 228 004 229 873.9	1.4	[16]
429 228 004 229 872.0	1.6	[17]
429 228 004 229 873.56	0.49	[18]
429 228 004 229 873.7	1.4	[19]
429 228 004 229 873.13	0.17	[20]
429 228 004 229 873.10	0.13	[21]
429 228 004 229 872.92	0.12	[22]

Frequency / Hz	Uncertainty / Hz	Reference
429 228 004 229 872.97	0.16	[23], 2014
429 228 004 229 873.04	0.11	[23], June 2015
429 228 004 229 872.97	0.40	[24]
429 228 004 229 872.99	0.18	[25]
$^{40}Ca^{+}$		
411 042 129 776 393.2	3.0 $^{(a)}$	[26]
411 042 129 776 398.4	1.2	[27]
411 042 129 776 400.5	1.2	[28], 2012
411 042 129 776 401.7	1.1	[28], 2014/2015
^{87}Rb		
6 834 682 610.904 312 9	2.4×10^{-6}	[29]
6 834 682 610.904 307 0	3.1×10^{-6}	[30]
6 834 682 610.904 312 5	2.1×10^{-6}	[31]
Frequency ratios	Fractional uncertainty	Reference
^{199}Hg /^{87}Sr		
2.629 314 209 898 909 60	8.4×10^{-17}	[32]
2.629 314 209 898 909 15	1.8×10^{-16}	[8]
^{199}Hg /^{87}Rb		
165 124.754 879 997 258	3.8×10^{-16}	[8]
^{171}Yb /^{87}Sr		
1.207 507 039 343 341 2	1.4×10^{-15}	[33], [34]
1.207 507 039 343 337 76	2.4×10^{-16}	[35]
1.207 507 039 343 337 749	4.6×10^{-17}	[36]
^{88}Sr /^{87}Sr		
1.000 000 144 883 682 777	$7 \times 10^{-17\,(b)}$	[37]
^{87}Sr/$^{40}Ca^{+}$		
1.044 243 334 529 641 6	2.3×10^{-15}	[27]
^{87}Sr/^{87}Rb		
62 801.453 800 512 435	3.3×10^{-16}	[22]

(a) From the least square procedure it turned out that the value and the uncertainty given in ref. [26] is not compatible with the remaining data. Thus the uncertainty given in the original publication [26] was increased to 3 Hz to make it statistically more consistent.

(b) The fractional uncertainty of 2.3×10^{-17} in ref. [37] has been increased by a factor of three since this uncertainty is an order of magnitude smaller than the other input data for this transition.

$^{40}Ca^{+}$, 4s $^{2}S_{1/2} - 3d\ ^{2}D_{5/2}$ unperturbed optical transition

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